NMC5344 Group Project

November 2026

1. Dataset Description

1.1 Dataset Overview

The dataset contains 89,619 observations and 47 variables, providing information on Steam games and their characteristics. These include game prices, downloadable content (DLC), achievements, operating system compatibility, estimated ownership, and peak concurrent players.

The dataset allows us to examine how game characteristics and platform accessibility are associated with player reach.

1.2 Variables and Data Types

The following table presents the key variables used in the analysis and their expected R data types.

Variable	R Data Type	Description
Owners	Character	Estimated ownership ranges
Peak players	Numeric	Highest concurrent player count
Price	Numeric	Game selling price
DLC count	Numeric	Number of DLC items
Achievements	Numeric	Number of achievements
Windows	Logical	Windows compatibility
Mac	Logical	macOS compatibility
Linux	Logical	Linux compatibility
Supported languages	Character	Available languages
Categories	Character	Game features
Genres	Character	Game genres
Tags	Character	Descriptive game tags
Developers	Character	Game developers
Packages	Character	Available game packages
Release date	POSIXct	Game release date

```
library(readxl)
library(dplyr)

games_data <- read_excel(
  '/Users/ken/Desktop/Coding/Group Project/archive/games_march2025_cleaned_2026_1105.xlsx'
)

games_data <- games_data[, 1:47]

dim(games_data)
```

```
## [1] 89624    47
```

```r
names(games_data)
```

```
##  [1] "appid"                    "name"
##  [3] "required_age"             "price"
##  [5] "dlc_count"                "windows"
##  [7] "mac"                      "linux"
##  [9] "metacritic_score"         "achievements"
## [11] "recommendations"          "supported_languages"
## [13] "full_audio_languages"     "developers"
## [15] "publishers"               "categories"
## [17] "genres"                   "score_rank"
## [19] "positive"                 "negative"
## [21] "estimated_owners"         "average_playtime_forever"
## [23] "average_playtime_2weeks"  "median_playtime_forever"
## [25] "median_playtime_2weeks"   "discount"
## [27] "peak_ccu"                 "pct_pos_total"
## [29] "num_reviews_total"        "pct_pos_recent"
## [31] "num_reviews_recent"       "...32"
## [33] "...33"                    "...34"
## [35] "...35"                    "...36"
## [37] "...37"                    "...38"
## [39] "...39"                    "...40"
## [41] "...41"                    "...42"
## [43] "...43"                    "...44"
## [45] "...45"                    "...46"
## [47] "...47"
```

```r
games_clean <- games_data %>%
  filter(
    !is.na(appid),
    !is.na(price),
    !is.na(dlc_count),
    !is.na(achievements),
    !is.na(peak_ccu)
  )

dim(games_clean)
```

```
## [1] 89614    47
```

2. Descriptive Statistics

2.1 Overall Descriptive Statistics

```r
library(tidyr)

overall_stats <- games_clean %>%
  summarise(
    across(
      c(price, dlc_count, achievements, peak_ccu),
      list(
        Mean = ~mean(.x, na.rm = TRUE),
```

Table 1: Overall Descriptive Statistics

Variable	Mean	Median	SD	Min	Max
price	7.31	4.99	13.33	0	999.98
dlc_count	0.60	0.00	15.35	0	3427.00
achievements	20.55	5.00	163.57	0	9821.00
peak_ccu	98.34	0.00	5717.67	0	1212356.00

```
        Median = ~median(.x, na.rm = TRUE),
        SD = ~sd(.x, na.rm = TRUE),
        Min = ~min(.x, na.rm = TRUE),
        Max = ~max(.x, na.rm = TRUE)
      )
    )
  ) %>%
  pivot_longer(
    everything(),
    names_to = c("Variable", ".value"),
    names_pattern = "^(.*)_(Mean|Median|SD|Min|Max)$"
  )


knitr::kable(
  overall_stats,
  digits = 2,
  format = "latex",
  caption = "Overall Descriptive Statistics",
  booktabs = TRUE,
  linesep = ""
)
```

```
library(haven)
library(readxl)
library(dplyr)
library(ggplot2)
library(tidyr)
library(stringr)
library(scales)

knitr::opts_chunk$set(message = FALSE, fig.width = 9, fig.height = 5.5)
theme_set(theme_minimal(base_size = 12) + theme(legend.position = "bottom"))
```

```
mac_stats <- games_clean %>%
  mutate(
    mac = toupper(as.character(mac))
  ) %>%
  filter(mac %in% c("TRUE", "FALSE")) %>%
  group_by(mac) %>%
  summarise(
    Games = n(),
    Mean = mean(peak_ccu, na.rm = TRUE),
    Median = median(peak_ccu, na.rm = TRUE),
```

```
    SD = sd(peak_ccu, na.rm = TRUE),
    Min = min(peak_ccu, na.rm = TRUE),
    Max = max(peak_ccu, na.rm = TRUE),
    .groups = "drop"
  )

knitr::kable(
  mac_stats,
  digits = 2,
  caption = "Peak Players by Mac Support",
  booktabs = TRUE
)
```

Table 2: Peak Players by Mac Support

mac	Games	Mean	Median	SD	Min	Max
FALSE	72176	90.38	0	5936.51	0	1212356
TRUE	17438	131.31	0	4704.85	0	555977

2.2 Descriptive Statistics by Mac Support

Descriptive statistics reveal substantial variation in game characteristics and player reach. The average game price is 7.31 (median = 4.99), while the average DLC count is 0.60 (median = 0). Similarly, the average number of achievements is 20.55 (median = 5). Peak concurrent players average 98.34, although the median is zero. These differences suggest right-skewed distributions, with relatively few games recording exceptionally high values.

Subgroup analysis shows that games supporting Mac have a higher mean peak player count (131.31) than games without Mac support (90.38). However, both groups have a median of zero, suggesting that unusually popular games influence the group averages. This indicates an association between platform compatibility and player reach rather than a causal relationship.

3. Visualization

```
owner_labels <- c("<20k", "20k-<100k", "100k-<500k", "500k-<2m", "2m+")
language_labels <- c("1 language", "2-5 languages", "6-10 languages", "11+ languages")

games <- games |>
  # Remove rows with misplaced text in the system columns.
  filter(if_all(c(windows, mac, linux), ~ toupper(.x) %in% c("TRUE", "FALSE"))) |>
  mutate(
    # Convert system support to 1/0, then add the three indicators.
    across(c(windows, mac, linux), ~ as.integer(toupper(.x) == "TRUE")),
    platform_count = windows + mac + linux,
    system_group = factor(platform_count, levels = 1:3,
                          labels = c("1 system", "2 systems", "3 systems")),

    # Count the quoted language names and place them into four groups.
    language_count = str_count(supported_languages, "'[^']+'"),
    language_group = cut(language_count, breaks = c(0, 1, 5, 10, Inf),
                         labels = language_labels),

    # Use the lower end of each ownership range to assign reach bands.
```

```
    owners_lower = as.numeric(str_extract(estimated_owners, "^\\d+")),
    ownership_group = cut(owners_lower,
                          breaks = c(0, 20000, 100000, 500000, 2000000, Inf),
                          right = FALSE, labels = owner_labels),
    broad_reach = owners_lower >= 100000,

    # Describe DLC both as availability and as four count categories.
    dlc_count = as.numeric(dlc_count),
    dlc_status = factor(if_else(dlc_count > 0, "Has DLC", "No DLC"),
                        levels = c("No DLC", "Has DLC")),
    dlc_group = cut(dlc_count, breaks = c(-1, 0, 1, 5, Inf),
                    labels = c("0", "1", "2-5", "6+"))
  ) |>
  # Use the same games in all four figures; exclude 0-0 ownership estimates.
  filter(estimated_owners != "0 - 0") |>
  drop_na(system_group, language_group, ownership_group, dlc_group)

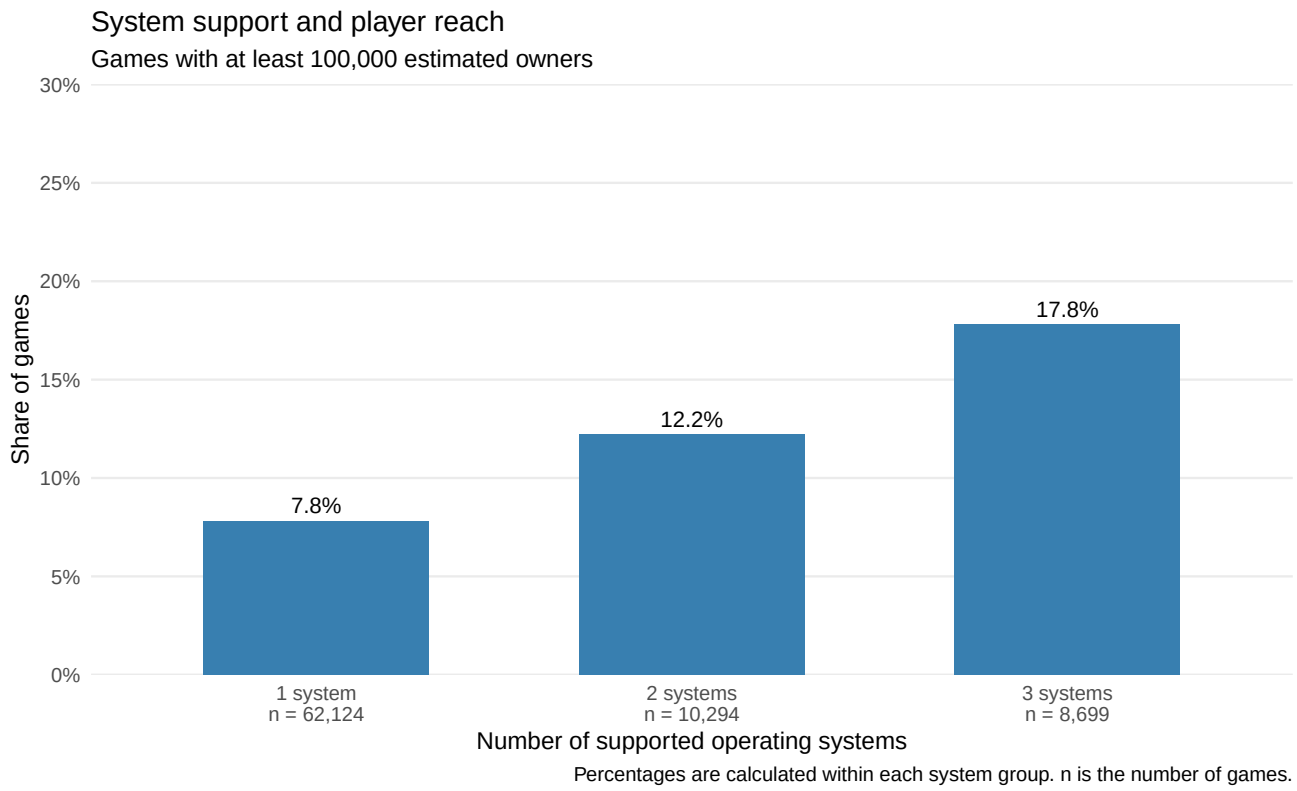
# Ownership-band colors for the DLC stacked bar chart.
owner_scale <- scale_fill_manual(
  values = c("#EEF3F8", "#C1D5E6", "#7EACCE", "#387FB0", "#124B78"),
  name = "Estimated owners", guide = guide_legend(nrow = 1)
)
```

The four figures use **81,117 games**. System and language coverage represent accessibility; DLC represents content expansion. Figures 1–2 and 4 compare the share reaching at least 100,000 estimated owners. Figure 3 shows the full ownership distribution.

3.1 Figure 1. System support and player reach

```
# Calculate group sizes and the percentage reaching 100,000 owners.
system_summary <- games |>
  group_by(system_group) |>
  summarise(n = n(), reach_100k = round(mean(broad_reach) * 100, 1))

# Each bar compares one group's percentage reaching 100,000 owners.
ggplot(system_summary, aes(x = system_group, y = reach_100k)) +
  geom_col(fill = "#387FB0", width = 0.6) +
  geom_text(aes(label = paste0(reach_100k, "%")), vjust = -0.5, size = 4) +
  scale_y_continuous(limits = c(0, 30), breaks = seq(0, 30, 5),
                     labels = label_number(suffix = "%"), expand = c(0, 0)) +
  scale_x_discrete(labels = paste0(system_summary$system_group,
                                   "\nn = ", comma(system_summary$n))) +
  labs(title = "System support and player reach",
       subtitle = "Games with at least 100,000 estimated owners",
       x = "Number of supported operating systems", y = "Share of games",
       caption = "Percentages are calculated within each system group. n is the number of games.") +
  theme(panel.grid.major.x = element_blank(), panel.grid.minor = element_blank())
```



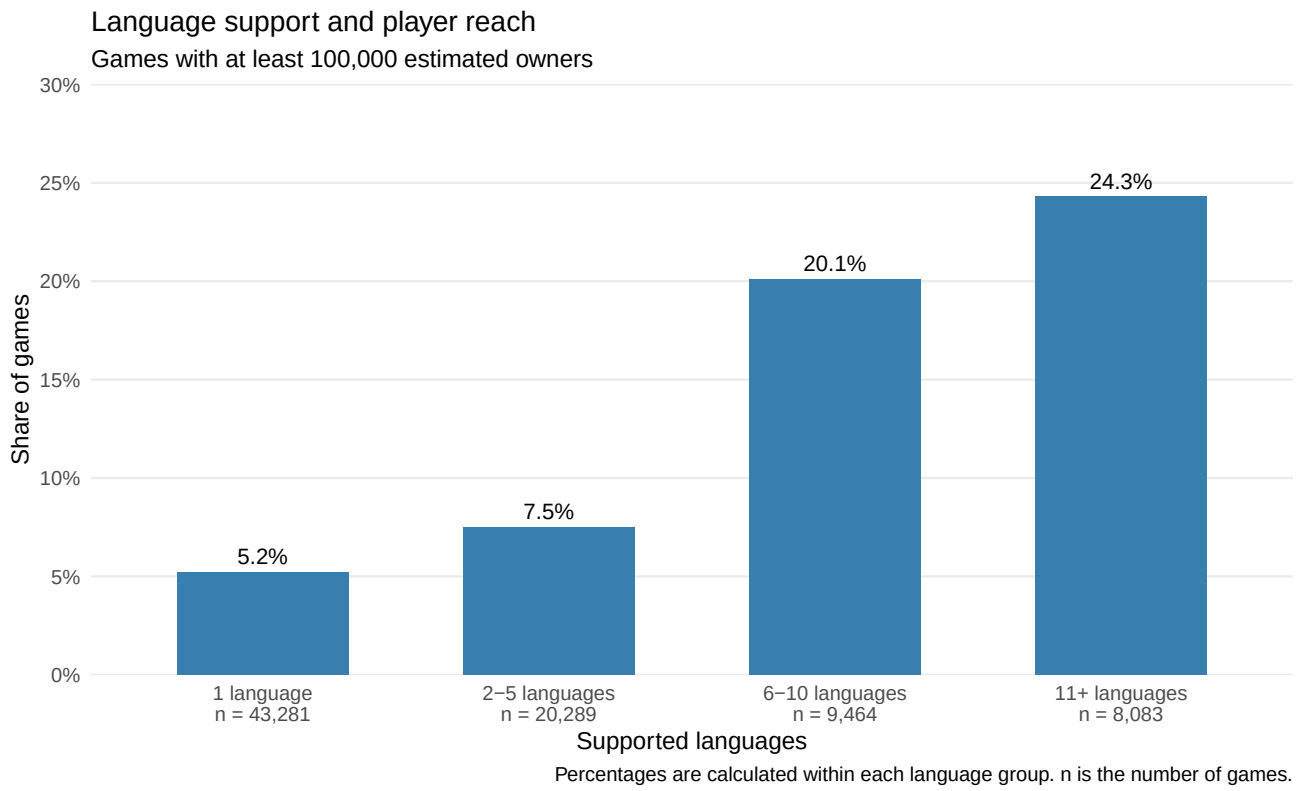
We use a bar chart to compare the share of games reaching at least 100,000 estimated owners across system-support groups. Focusing on this threshold makes the relationship clearer without the larger under-100,000 group dominating the display.

Broader system coverage is associated with wider player reach. The share reaching at least 100,000 estimated owners rises from **7.8%** for one system to **12.2%** for two systems and **17.8%** for three systems. The three-system group has more than twice the proportion reaching this threshold as the one-system group, linking broader technical accessibility to reach.

3.2 Figure 2. Language support and player reach

```
language_summary <- games |>
  group_by(language_group) |>
  summarise(n = n(), reach_100k = round(mean(broad_reach) * 100, 1))

# Keep the same 0-30% vertical axis as Figure 1 for direct comparison.
ggplot(language_summary, aes(x = language_group, y = reach_100k)) +
  geom_col(fill = "#387FB0", width = 0.6) +
  geom_text(aes(label = paste0(reach_100k, "%")), vjust = -0.5, size = 4) +
  scale_y_continuous(limits = c(0, 30), breaks = seq(0, 30, 5),
                     labels = label_number(suffix = "%"), expand = c(0, 0)) +
  scale_x_discrete(labels = paste0(language_summary$language_group,
                                   "\nn = ", comma(language_summary$n))) +
  labs(title = "Language support and player reach",
       subtitle = "Games with at least 100,000 estimated owners",
       x = "Supported languages", y = "Share of games",
       caption = "Percentages are calculated within each language group. n is the number of games.") +
  theme(panel.grid.major.x = element_blank(), panel.grid.minor = element_blank())
```



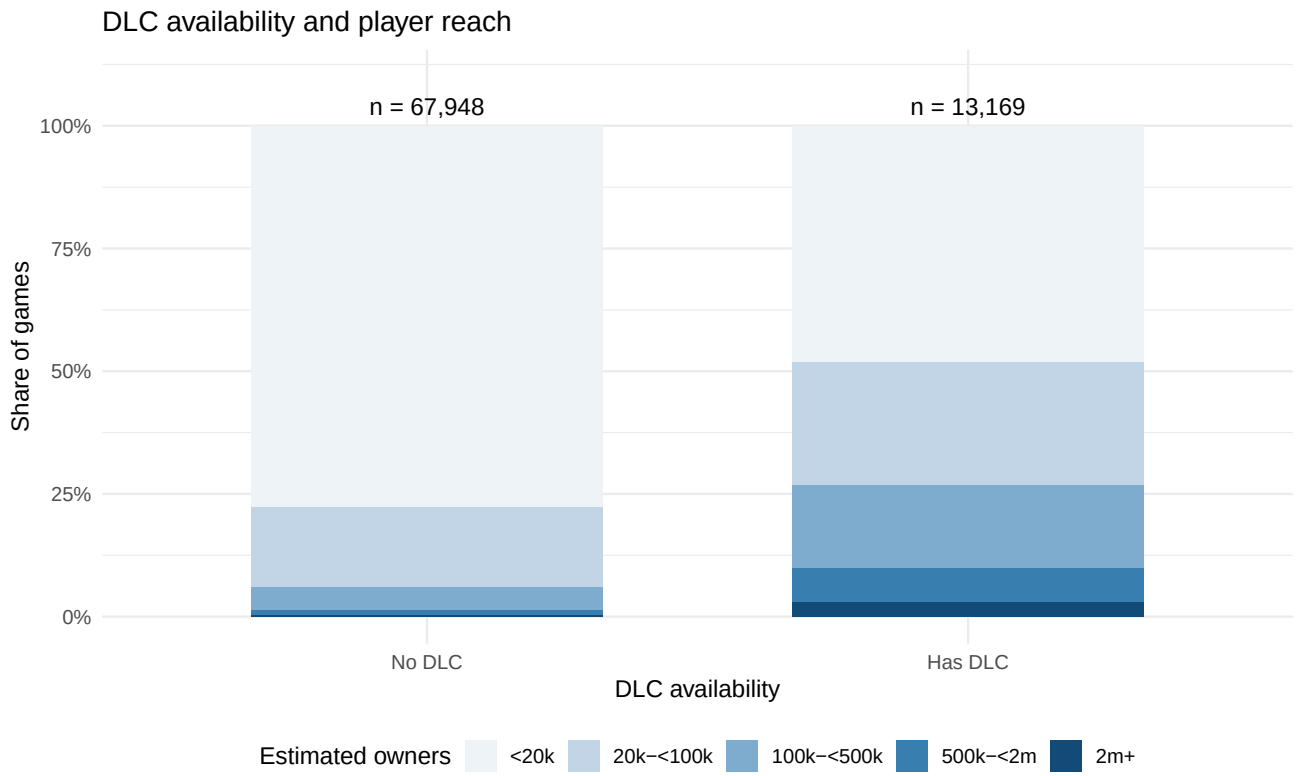
To show how language coverage relates to wider reach, we plot the percentage of games with at least 100,000 estimated owners. This keeps the comparison clear despite the concentration of games below that threshold.

Broader language coverage is associated with wider player reach, particularly in the 6–10 and 11+ language groups. The share reaching at least 100,000 estimated owners increases across the four language groups: **5.2%**, **7.5%**, **20.1%** and **24.3%**. Together, Figures 1–2 show that broader technical and language accessibility both accompany a larger share of games with wider reach.

3.3 Figure 3. DLC availability and player reach

```
dlc_summary <- games |>
  group_by(dlc_status) |>
  summarise(n = n(), reach_100k = round(mean(broad_reach) * 100, 1))

# Compare ownership within games with and without DLC.
ggplot(games, aes(x = dlc_status, fill = ownership_group)) +
  geom_bar(position = "fill", width = 0.65) +
  geom_text(data = dlc_summary,
            aes(x = dlc_status, y = 1.04, label = paste0("n = ", comma(n))),
            inherit.aes = FALSE) +
  scale_y_continuous(labels = label_percent(), breaks = seq(0, 1, 0.25),
                     limits = c(0, 1.1)) +
  owner_scale +
  labs(title = "DLC availability and player reach",
       x = "DLC availability", y = "Share of games")
```



A 100% stacked bar chart allows us to compare the full ownership distributions of games with and without DLC. Using percentages makes these groups comparable despite their different sizes.

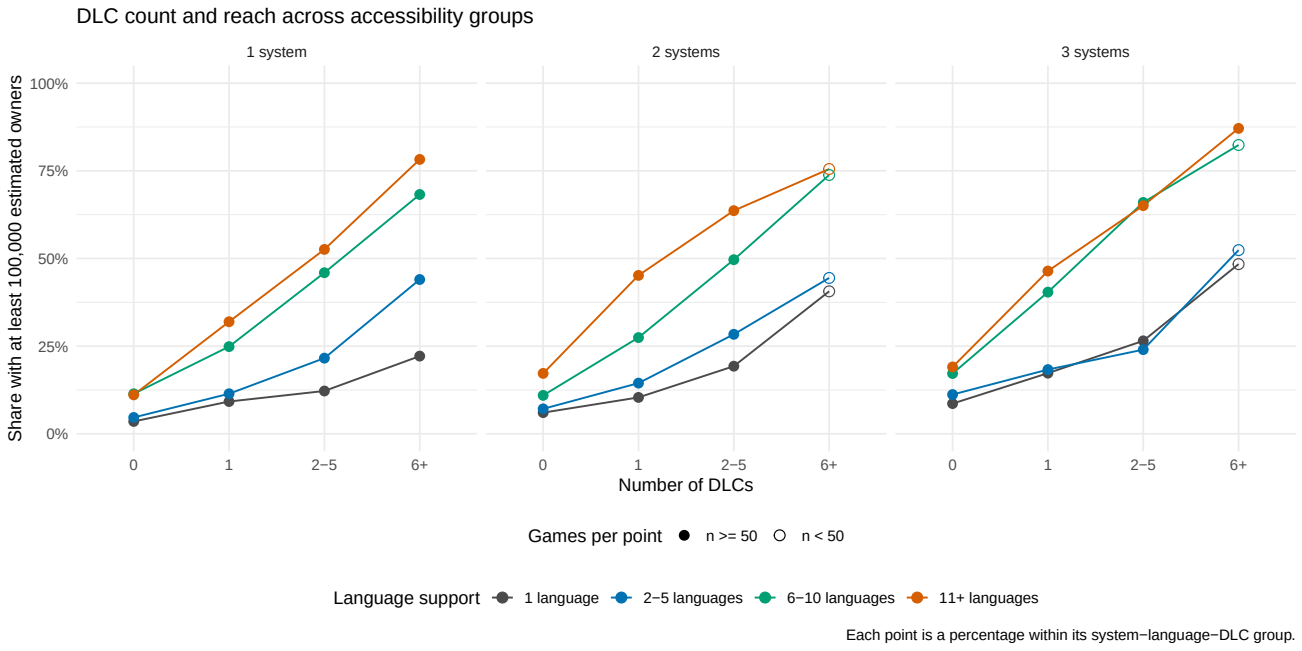
DLC availability is positively associated with player reach. Among games without DLC, **6%** reach at least 100,000 estimated owners, compared with **26.8%** of games with DLC. Games offering additional content are therefore more often represented in the broader-reach bands. Alongside Figures 1–2, this suggests that both accessibility and content expansion characterize games with wider reach. Figure 4 examines how these features relate to reach when considered together.

3.4 Figure 4. DLC count and reach across accessibility groups

```
# Each row represents a system-language-DLC combination. The mean of
# TRUE/FALSE gives the share of games reaching 100,000 owners in that group.
combined <- games |>
  group_by(system_group, language_group, dlc_group) |>
  summarise(n = n(), share = mean(broad_reach), .groups = "drop")

# Panels separate system coverage; colors separate language coverage.
# Hollow points identify combinations containing fewer than 50 games.
ggplot(combined, aes(x = dlc_group, y = share,
                     color = language_group, group = language_group)) +
  geom_line(linewidth = 0.6) +
  geom_point(aes(shape = n < 50), size = 3) +
  facet_wrap(~ system_group, nrow = 1) +
  scale_color_manual(values = c("#4B4B4B", "#0072B2", "#009E73", "#D55E00"),
                     name = "Language support") +
  scale_shape_manual(values = c("FALSE" = 16, "TRUE" = 1),
                     labels = c("n  50", "n < 50"), name = "Games per point") +
  scale_y_continuous(labels = label_percent(), limits = c(0, 1)) +
```

```
  labs(title = "DLC count and reach across accessibility groups",
       x = "Number of DLCs", y = "Share with at least 100,000 estimated owners",
       caption = "Each point is a percentage within its system-language-DLC group.") +
  theme(legend.box = "vertical")
```



We bring the three features together in a faceted point-and-line chart to examine whether the DLC–reach pattern differs across accessibility groups. Separate panels show system coverage, while colors distinguish language coverage.

More DLC and broader accessibility are jointly associated with wider player reach. The share reaching 100,000 estimated owners increases across DLC-count categories in all 12 system–language groups, so the positive DLC pattern remains visible within each accessibility profile. At the same DLC level, games with broader language coverage generally have higher reach proportions; broader system coverage also generally corresponds to higher proportions across panels, with some subgroup exceptions. For example, among one-system games supporting 11+ languages, the share rises from **11.1%** with no DLC to **78.2%** with 6+ DLCs, compared with **3.5%** to **22.2%** for one-language games. This illustrates how the DLC–reach pattern varies with language coverage: games combining extensive content expansion with broader language access have particularly high reach proportions. Seven hollow points have fewer than 50 games, so those percentages are less stable.

Across the four figures, broader system support, broader language support and DLC availability are associated with wider player reach. The combined figure also shows variation between accessibility groups. The subsequent modelling analysis will further examine these associations.

4. Regression Analysis

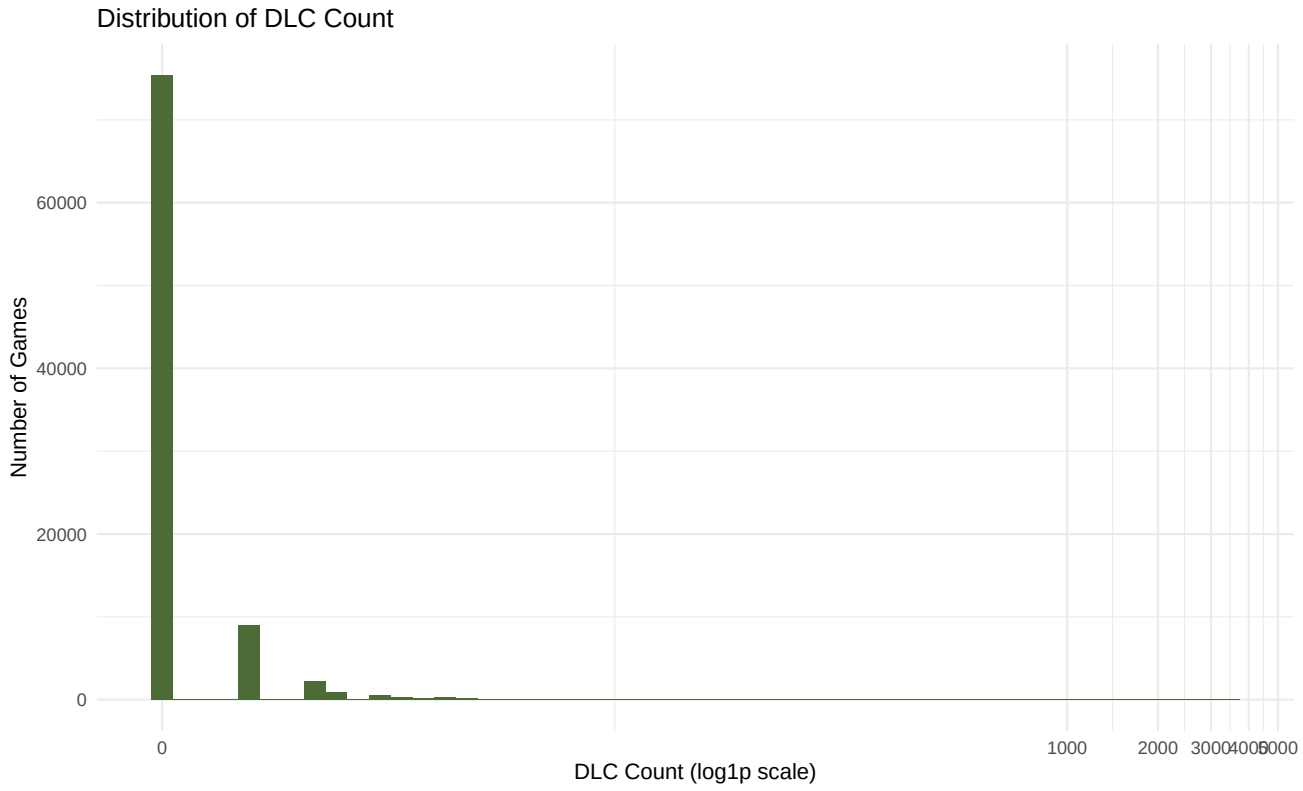
4.1 Data Preparation for Regression

```
games_clean <- dataset_1 %>%
  mutate(
    dlc_log = log1p(dlc_count)
  )
```

```
ggplot(games_clean, aes(x = dlc_count)) +
  geom_histogram(bins = 50, fill = "#4C6B37") +
  scale_x_continuous(trans = "log1p") +
  labs(
    title = "Distribution of DLC Count",
    x = "DLC Count (log1p scale)",
    y = "Number of Games"
  ) +
  theme_minimal()
```

4.1.1 DLC Count Distribution and Log Transformation

```
## Warning: Removed 6 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
games_clean <- games_clean %>%
  mutate(
    windows = as.integer(dataset_1$windows),
    mac = as.integer(dataset_1$mac),
    linux = as.integer(dataset_1$linux)
  )
games_clean <- games_clean %>%
  mutate(
    OS_type = case_when(
      is.na(windows) | is.na(mac) | is.na(linux) ~ NA_character_,
      windows == 1 & mac == 0 & linux == 0 ~ "Windows Only",
```

```
      windows == 1 & mac == 1 & linux == 0 ~ "Windows + Mac",
      windows == 1 & mac == 0 & linux == 1 ~ "Windows + Linux",
      windows == 1 & mac == 1 & linux == 1 ~ "Windows + Mac + Linux",
      windows == 0 & mac == 1 & linux == 0 ~ "Mac Only",
      windows == 0 & mac == 0 & linux == 1 ~ "Linux Only",
      windows == 0 & mac == 1 & linux == 1 ~ "Mac + Linux",
      windows == 0 & mac == 0 & linux == 0 ~ "No Supported OS"
    )
  )
games_clean %>%
  count(OS_type, sort = TRUE)
```

4.1.2 Classification of Operating System (OS) Support

```
## # A tibble: 8 x 2
##   OS_type                   n
##   <chr>                 <int>
## 1 Windows Only          68927
## 2 Windows + Mac + Linux  9377
## 3 Windows + Mac          8039
## 4 Windows + Linux        3241
## 5 Mac Only                 21
## 6 <NA>                     10
## 7 Linux Only                8
## 8 Mac + Linux               1
```

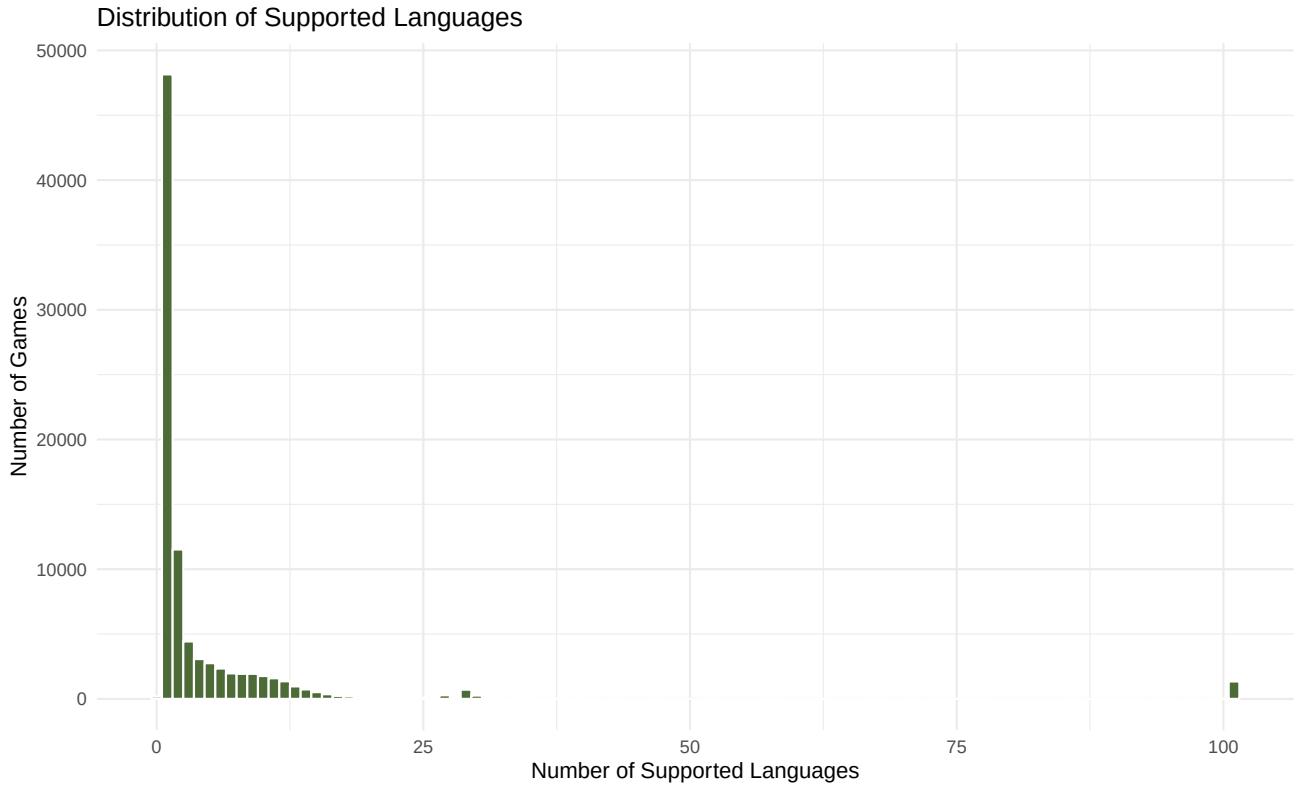
```
library(stringr)

games_clean <- games_clean %>%
  mutate(
    language_count = case_when(
      is.na(supported_languages) ~ NA_integer_,
      str_trim(supported_languages) == "[]" ~ 0L,
      str_detect(
        supported_languages,
        "^\\[.*\\]$"
      ) ~ str_count(
        supported_languages,
        "',\\s*'"
      ) + 1L,
      TRUE ~ NA_integer_
    )
  )
ggplot(games_clean, aes(x = language_count)) +
  geom_histogram(
    binwidth = 1,
    fill = "#4C6B37",
    color = "white"
  ) +
  labs(
    title = "Distribution of Supported Languages",
    x = "Number of Supported Languages",
```

```
    y = "Number of Games"
  ) +
  theme_minimal()
```

4.1.3 Number of Supported Languages

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
games_clean <- games_clean %>%
  mutate(
    zero_owner_range = estimated_owners == "0 - 0"
  )

games_clean <- games_clean %>%
   select(
    -any_of(c(
      "zero_owner_range",
      "owner_range_valid",
      "owners_lower",
      "owners_upper",
      "owner_midpoint",
      "owners_log"
    ))
  ) %>%
  mutate(
```

```
    owner_range_valid = str_detect(
      estimated_owners,
      "^\\s*[0-9]+\\s*-\\s*[0-9]+\\s*$"
    )
  ) %>%
  separate_wider_delim(
    estimated_owners,
    delim = " - ",
    names = c("owners_lower", "owners_upper"),
    cols_remove = FALSE,
    too_few = "align_start"
  ) %>%
  mutate(
    owners_lower = as.numeric(owners_lower),
    owners_upper = as.numeric(owners_upper),
    owner_midpoint = if_else(
      owner_range_valid &
        owners_lower <= owners_upper,
      (owners_lower + owners_upper) / 2,
      NA_real_
    ),
    owners_log = log1p(owner_midpoint)
  )
```

4.1.4 Transformation of Estimated Ownership Ranges

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `owners_lower = as.numeric(owners_lower)`.
## Caused by warning:
## ! NAs introduced by coercion
```

```
library(dplyr)

model_data <- games_clean %>%
  select(
    appid,
    name,
    owners_log,
    dlc_log,
    OS_type,
    language_count
  ) %>%
  filter(
    OS_type %in% c(
      "Windows Only",
      "Windows + Mac",
      "Windows + Linux",
      "Windows + Mac + Linux"
    )
  ) %>%
  mutate(
    OS_type = factor(OS_type),
```

```
    OS_type = relevel(
      OS_type,
      ref = "Windows Only"
    )
  )

glimpse(model_data)
```

4.1.5 Selection of Regression Sample

```
## Rows: 89,584
## Columns: 6
## $ appid          <dbl> 730, 578080, 570, 271590, 359550, 440, 105600, 252490, ~
## $ name           <chr> "Counter-Strike 2", "PUBG: BATTLEGROUNDS", "Dota 2", "G~
## $ owners_log     <dbl> 18.82615, 18.13300, 19.67344, 18.13300, 17.37086, 17.37~
## $ dlc_log        <dbl> 0.6931472, 0.0000000, 1.0986123, 0.0000000, 2.3025851, ~
## $ OS_type        <fct> Windows + Linux, Windows Only, Windows + Mac + Linux, W~
## $ language_count <int> 29, 17, 28, 13, 16, 27, 9, 25, 25, 14, 13, 17, 15, 12, ~
```

4.2 Model Specification

Since the purpose of the project is to find the potential variables that may be associated with the number of estimated game owners, this project chose three variables relating to game design and hopes to provide some suggestions for future game design on STEAM. Based on the purpose, our group selects 4 different linear models. Overall, four different models examine the main and interactive effects of these three variables on the number of estimated owners. Therefore, the dependent variable is the number of estimated owners, and the independent variables are the number of DLC, supported language, and the types of OS. The detailed information is shown below.

```
m1 <- lm(
  owners_log ~ dlc_log + OS_type + language_count,
  data = model_data
)
```

4.2.1 Model 1 — Baseline Model

```
m2 <- lm(
   owners_log ~ dlc_log*OS_type + language_count,
  data = model_data
)
```

4.2.2 Model 2 — DLC × Operating System Interaction

```
m3 <- lm(
   owners_log ~ dlc_log*language_count + OS_type,
  data = model_data
)
```

4.2.3 Model 3 — DLC × Language Support Interaction

```
m4 <- lm(
   owners_log ~ dlc_log+ OS_type*language_count,
  data = model_data
)
```

```
summary(m1)
```

4.2.4 Model 4 — Operating System × Language Support Interaction

```
##
## Call:
## lm(formula = owners_log ~ dlc_log + OS_type + language_count,
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5194   0.2950   0.4107   0.9637  10.4847
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  8.8392576  0.0124216 711.603   <2e-16 ***
## dlc_log                      0.9937357  0.0222339  44.695   <2e-16 ***
## OS_typeWindows + Linux      -0.0411802  0.0538222  -0.765    0.444
## OS_typeWindows + Mac         0.3526875  0.0353219   9.985   <2e-16 ***
## OS_typeWindows + Mac + Linux 0.6369586  0.0330882  19.250   <2e-16 ***
## language_count              -0.0394984  0.0007682 -51.420   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.994 on 89578 degrees of freedom
## Multiple R-squared:  0.05279,    Adjusted R-squared:  0.05273
## F-statistic: 998.4 on 5 and 89578 DF,  p-value: < 2.2e-16
```

```
summary(m2)
```

```
##
## Call:
## lm(formula = owners_log ~ dlc_log * OS_type + language_count,
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.9513   0.2773   0.4048   0.9571  10.5610
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         8.8450976  0.0125722 703.546  < 2e-16 ***
## dlc_log                             0.9515743  0.0261912  36.332  < 2e-16 ***
```

```
## OS_typeWindows + Linux               -0.0207018  0.0586400  -0.353  0.72407
## OS_typeWindows + Mac                  0.3247765  0.0382173   8.498  < 2e-16 ***
## OS_typeWindows + Mac + Linux          0.5869752  0.0366936  15.997  < 2e-16 ***
## language_count                       -0.0394529  0.0007683 -51.354  < 2e-16 ***
## dlc_log:OS_typeWindows + Linux       -0.1078345  0.1325023  -0.814  0.41574
## dlc_log:OS_typeWindows + Mac          0.1454631  0.0719614   2.021  0.04324 *
## dlc_log:OS_typeWindows + Mac + Linux  0.2086291  0.0648161   3.219  0.00129 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.994 on 89575 degrees of freedom
## Multiple R-squared:  0.05294,    Adjusted R-squared:  0.05285
## F-statistic: 625.8 on 8 and 89575 DF,  p-value: < 2.2e-16
```

```
summary(m3)
```

```
##
## Call:
## lm(formula = owners_log ~ dlc_log * language_count + OS_type,
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.3208   0.2970   0.3957   0.9954  10.5374
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   8.857577   0.012471  710.25   <2e-16 ***
## dlc_log                       0.765367   0.027218   28.12   <2e-16 ***
## language_count               -0.042839   0.000801  -53.48   <2e-16 ***
## OS_typeWindows + Linux       -0.037629   0.053760   -0.70    0.484
## OS_typeWindows + Mac          0.355732   0.035281   10.08   <2e-16 ***
## OS_typeWindows + Mac + Linux  0.646608   0.033056   19.56   <2e-16 ***
## dlc_log:language_count        0.031388   0.002163   14.51   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.991 on 89577 degrees of freedom
## Multiple R-squared:  0.05501,    Adjusted R-squared:  0.05494
## F-statistic:   869 on 6 and 89577 DF,  p-value: < 2.2e-16
```

```
summary(m4)
```

```
##
## Call:
## lm(formula = owners_log ~ dlc_log + OS_type * language_count,
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.3459   0.2176   0.4183   0.8919  10.6960
##
## Coefficients:
##                                          Estimate Std. Error t value
```

```
## (Intercept)                                  8.8294189  0.0125715 702.338
## dlc_log                                      0.9846045  0.0222153  44.321
## OS_typeWindows + Linux                      -0.0847237  0.0579311  -1.462
## OS_typeWindows + Mac                         0.1920186  0.0403615   4.757
## OS_typeWindows + Mac + Linux                 0.8038875  0.0359828  22.341
## language_count                              -0.0372880  0.0008685 -42.932
## OS_typeWindows + Linux:language_count        0.0086931  0.0042810   2.031
## OS_typeWindows + Mac:language_count          0.0341148  0.0041253   8.270
## OS_typeWindows + Mac + Linux:language_count -0.0247693  0.0021602 -11.466
##                                                Pr(>|t|)
## (Intercept)                                   < 2e-16 ***
## dlc_log                                       < 2e-16 ***
## OS_typeWindows + Linux                         0.1436
## OS_typeWindows + Mac                         1.96e-06 ***
## OS_typeWindows + Mac + Linux                  < 2e-16 ***
## language_count                                < 2e-16 ***
## OS_typeWindows + Linux:language_count          0.0423 *
## OS_typeWindows + Mac:language_count           < 2e-16 ***
## OS_typeWindows + Mac + Linux:language_count   < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.991 on 89575 degrees of freedom
## Multiple R-squared:  0.05515,    Adjusted R-squared:  0.05507
## F-statistic: 653.6 on 8 and 89575 DF,  p-value: < 2.2e-16
```

4.3 Model Comparison

```
# Windows Only
coef(m4)["language_count"]
```

```
## language_count
##    -0.03728803
```

```
# Windows + Linux
coef(m4)["language_count"] +
  coef(m4)["OS_typeWindows + Linux:language_count"]
```

```
## language_count
##    -0.02859497
```

```
# Windows + Mac
coef(m4)["language_count"] +
  coef(m4)["OS_typeWindows + Mac:language_count"]
```

```
## language_count
##   -0.003173259
```

```
# Windows + Mac + Linux
coef(m4)["language_count"] +
  coef(m4)["OS_typeWindows + Mac + Linux:language_count"]
```

```
## language_count
##     -0.0620573
```

```
round(
  sapply(
  list(M1 = m1, M2 = m2, M3 = m3, M4 = m4),
  function(m) summary(m)$adj.r.squared
),
4)
```

```
##     M1     M2     M3     M4
## 0.0527 0.0529 0.0549 0.0551
```

These 4 models can be compared based on the changes in adjusted R-square. According to the adjusted R-square, models 3 and 4 are better, since the values of their adjusted R-square are the highest two among the four models. Therefore, the following interpretation mainly focuses on these two models.

4.4 Results and Interpretation

As for model 3, there is an effect of the number of languages on the relationship between the number of estimated owners and DLC ($p<0.001$, B= 0.031). Specifically, the association between the number of estimated owners and DLC will become stronger when the number of supporting languages increases. In other words, the number of languages is a moderator between the number of estimated owners and DLC. As for model 4, different OS is also a moderator of the negative relationship between the number of supported languages and estimated owners. However, such moderating effects vary. Compared with Windows-only OS (B=-0.037), games that support Windows + Linux OS ($p<0.05$, B=-0.029) and Windows + Mac OS($p<0.001$, B=-0.003) weaken the negative association. Nonetheless, the negative relationship between the number of supported languages and estimated owners is strengthened with games supporting Windows + Linux+ Mac OS ($p<0.001$, B=-0.062).